

NGUYEN TUAN PHAT

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Objective

AI-oriented Financial Technology undergraduate with project-based experience in machine learning model development, data preprocessing, and experimental evaluation. Experienced in using Python to process and analyze datasets, build predictive models, and extract insights from structured and unstructured data. Particularly interested in applying AI and data analytics to real-world business and financial problems, with a long-term goal of developing expertise in applied AI and data-driven decision systems.

Skills & Interests

Machine Learning & AI

- Model evaluation (Accuracy, Precision, Recall, F1-score)
- Feature engineering & cross-validation
- Vector embeddings & similarity search (RAG systems)
- Basic understanding of recommendation systems
- LLM applications (LangChain, HuggingFace)

Programming & Data Processing

- Python (Pandas, NumPy, Scikit-learn)
- SQL (JOIN, GROUP BY, data querying)
- Data Cleaning, Transformation, and Preprocessing

Data Visualization & Tools

- PostgreSQL, MySQL
- Power BI, Excel (Data Visualization, Interactive Dashboards)
- GitHub (version control)

Soft skills:

- Team leadership & project planning (academic project leader)
 - Strong self-learning ability
 - Analytical problem solving
 - Communication and presentation skills
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Project

LLM RAG Assistant

April 2026

[nguyentuanphat2594/LLM-RAG](#)

Tools: Python, LangChain, HuggingFace, ChromaDB, Streamlit

- Implemented a Retrieval-Augmented Generation (RAG) system for question answering over PDF documents based on existing architecture
- Integrated LangChain pipeline including document loading, semantic chunking, embedding, retrieval, and LLM generation
- Used HuggingFace embedding model and open-source LLM for text generation
- Connected ChromaDB vector database for similarity search over document chunks

AI Cocktail Recommendation

May 2026

[nguyentuanphat2594/AI-Cocktail-Recommendation-System](#)

Tools: Python, PostgreSQL, pgvector, Sentence Transformers, Streamlit

- Modified an existing vector-based recommendation system for cocktail retrieval
- Added alcohol preference filtering for ingredient/style/occasion searches
- Built a dashboard to analyze ingredient usage and cocktail trends
- Debugged database and Streamlit session-state issues

Crypto Market Analysis & Modeling Dashboard ([Click to streamlit website](#))

Dec 2025

[nguyentuanphat2594/Crypto_Analysis](#)

Tools: Pandas, Seaborn, Scikit-learn

- Performed time-series exploratory data analysis on market datasets
- Built a machine learning pipeline including feature engineering, model training, hyperparameter tuning, and walk-forward validation for cryptocurrency trend prediction.
- Developed and evaluated a Random Forest-based trading model with hyperparameter tuning and walk-forward optimization, reporting train/test performance.
- Achieved ~19% CAGR over 5+ years with walk-forward optimization, Sharpe Ratio > 2.0, Max Drawdown < 23%, 65%

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win rate, and Profit Factor > 3 on out-of-sample test data.

Student Management System

[nguyentuanphat2594/Student-Management-System](#)

Tools: SQL Server

- Designed and implemented a relational database system simulating a university environment
- Developed SQL queries (JOIN, GROUP BY, subqueries) to analyze student distribution, course registration, and academic performance
- Built queries for GPA calculation and course enrollment analysis
- Designed ERD and database schema, ensuring data consistency across tables

Education

HCMC University of Banking

Bachelor, Financial Technology.

GPA: 3.2/4.0

Thu Duc Campus

2022 – May 2026

Certifications

English (VSTEP B2), Microsoft Office Specialist (MOS)

Relevant Coursework:

Data Mining, Data Warehouse & Decision Support Systems,
Machine Learning, Database Systems, Financial Analysis